

KAN-Based Adaptive Parameter Control for Multi-User Satellite FSO/QKD Systems

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Abstract—We address adaptive parameter control for multi-user satellite free-space-optical (FSO) quantum key distribution (QKD) using subcarrier-intensity-modulation BPSK with dual-threshold direct detection (SIM-BPSK/DT-DD). Extending the geostationary/low-Earth-orbit (GEO/LEO) multi-user architecture, we formulate the joint optimization of the modulation depth, the Alice and per-user dual-threshold coefficients, and the inter-user exclusion ratio as a constrained nonconvex problem that couples all N Alice-Bob pairs through a global modulation depth, a closed-form asymmetric exclusion field, and the unauthorized-receiver (URA) and beam-splitting (BSA) security constraints. We solve it by a two-level decomposition, an outer search over the global variables wrapping a mean-field block-coordinate inner solver over the thresholds, whose outer/inner split is exact and which serves as the oracle for a learned two-stage controller. We compare Kolmogorov–Arnold Networks (KAN), regularized multi-layer perceptrons (MLP), and a linear baseline on a five-axis trade-off (accuracy, closed-loop secret-key retention, parameter count, latency, interpretability). Validated against an SGP4-propagated Starlink-density Walker constellation over a Hanoi ground station, the controller delivers a $2.10\times$ secret-key gain and a $3.6\times$ secure pair-time improvement (18 vs 5 secure pair-steps, $N=3$ heterogeneous users); on a sparser shell no static parameter choice is secure at all, making adaptation a precondition for operation. KAN uses $3.3\times$ fewer parameters than the MLP at matching key retention and yields a closed-form rule for the dual-threshold coefficient with $R^2 \approx 0.97$.

Reproducibility. All code, bundled TLE data, and run scripts that reproduce every figure and table in this paper are available at <https://github.com/haodpsut/kan-satellite-fso-qkd>; the head-line runs reproduce in under one hour on a single CPU plus an RTX-class GPU for the KAN training phase.

Index Terms—Quantum key distribution, free-space optical communications, satellite networks, Kolmogorov–Arnold Networks, multi-user QKD, adaptive control.

I. INTRODUCTION

SATELLITE-based free-space optical (FSO) channels are the most credible medium for intercontinental quantum key distribution (QKD): they avoid the exponential photon-loss bound of optical fiber and exploit the near-vacuum of space for low excess noise [1], [2], [3], [4]. The 2017 demonstration of decoy-state BB84 between the Micius satellite and ground stations in China [1], the simultaneous entanglement distribution at 1,200 km baseline [5], and the relayed intercontinental key exchange of [6] together establish satellite QKD as a near-term

technology rather than a research curiosity. Recent progress is summarized in [3], [4].

Among the QKD families adapted to FSO, the subcarrier-intensity-modulation BPSK with dual-threshold direct detection (SIM-BPSK/DT-DD) scheme proposed by Vu, Pham, Dang, Le, and Pham [7], [8], [9] is attractive because it relies on entirely classical components (an EDFA-amplified GEO source, PIN photodetectors, and dual-threshold post-processing), yet achieves information-theoretic security against unauthorized-receiver attacks (URA) and beam-splitting attacks (BSA) [10], [11], provided that the modulation depth μ , the DT scale coefficients β_A, β_i , and the multi-user fading-randomness exclusion ratio χ are appropriately chosen. SIM-BPSK is well established for classical FSO links [12], [13], and DT-DD inherits its noise-budget structure from the standard outage-capacity analysis [14], [15].

The three papers in the lineage establish, in increasing generality, the relay-assisted single-link case [7], the entanglement-based two-user case [8] (which adapts the BBM92 protocol [16] to the DT/DD measurement basis), and the GEO source serving N ground users via dual LEO relays with a closed-form inclusion-exclusion term [9]. In all three, the parameters (μ, β, χ) are picked *statically*: a single operating point is selected by reading the analytic performance plots at a representative geometry, and that point is then held throughout operation.

We show in this paper that for the realistic case of a LEO relay traversing its pass while serving an asymmetric cluster of ground users, the static-parameter assumption is the dominant performance bottleneck. Peak elevation, slant range, log-amplitude turbulence variance [17], [18], and the URA leakage geometry vary by orders of magnitude across a single 5 to 10 minute pass, and the optimal $\mathbf{p}(t) = (\mu, \beta_A, \{\beta_i\}, \chi)$ drifts continuously with the time-varying state $\mathbf{s}(t)$. Under realistic constellations propagated from real Two-Line Elements (TLE) via the SGP4 propagator [19], [20] on Starlink-density shells [21], [22], we further show that *no* single static $\mathbf{p}$ produces any secure pair-step over a typical 90-minute observation window, so adaptation is a precondition for the system to operate at all rather than a marginal improvement.

A. Contributions

This paper makes the following contributions:

- 1) **Full multi-user joint problem.** We restate the GEO/multi-LEO/ N -user system of [9] together with the pointing-error/Gaussian-beam model of [7] (dropped in [9]) and derive the corresponding joint problem (**P**) over

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$\mathbf{p} = (\mu, \beta_A, \{\beta_i\}, \chi)$ with URA, BSA, and an inter-user secrecy constraint. The problem is nonconvex with three sources of coupling: μ shared globally, χ entering every pair through a product field $\phi_i = \prod_{j \neq i} (1 - P_c^j)$, and β_A shared between Alice and every Bob. We generalize the inclusion–exclusion exclusion term of [9, Eq. 22] to asymmetric clusters in closed form, recovering the symmetric case as a special case.

- 2) **Principled two-level decomposition.** We expose the variable roles to obtain a two-level decomposition whose outer/inner split is exact (a relaxation-free reformulation of **(P)**): an outer two-dimensional search over (μ, χ) wraps an inner mean-field block-coordinate solver over the thresholds. The inner solver iterates a 1-D maximization for each β_i given an exclusion field and Alice's β_A , then refreshes the field. We prove that the symmetric-cluster fixed point recovers the closed form of [9, Eq. 24] and we numerically verify that the decomposition equals a brute-force joint grid optimum on a two-user cluster to machine precision.
- 3) **Two-stage controller with KAN.** The decomposed oracle generates supervised pairs $(s(t) \rightarrow p^*(t))$ for a two-stage controller: a global head consumes cluster-aggregate features and predicts (μ, χ, β_A) ; a per-user shared head consumes $(s_i, \beta_A, \mu, \chi)$ and predicts β_i . We compare a Kolmogorov-Arnold Network (KAN) [23], [24] against a regularized multi-layer perceptron and a linear residual baseline on a five-axis trade-off (regression accuracy, closed-loop secret-key retention, parameter count, inference latency, and interpretability). Deep learning has been widely studied for the classical physical layer [25], [26], [27] and adaptive control via reinforcement learning [28]; KAN is attractive for the QKD setting specifically because its trained spline activations admit symbolic extraction [24], [29], which we use to produce a closed-form rule for β^* .
- 4) **TLE-driven realistic-orbit validation.** We implement an SGP4 propagator path mirroring the analytic-orbit API so that every script accepts `-tle` as a drop-in switch. A synthetic Walker constellation (P planes $\times S$ satellites/plane) reproduces Starlink-shell coverage statistics in the absence of internet access; a real TLE bundle from Celestrak can be loaded with `-tle PATH`. Over a Hanoi ground station, the static baseline collapses to zero secure pair-steps on the Walker constellation, while the adaptive controller recovers a non-trivial fraction near overhead.
- 5) **Honest multi-axis evaluation.** We report results with explicit mean $\pm$ std over five seeds, fair (regularized) MLP baselines, and a model-agnostic β safety-margin recipe that lifts feasibility by 1.1–1.5 $\times$ for every controller. KAN does not dominate every axis; its value is parameter efficiency plus an interpretable closed-form rule for β with symbolic $R^2 \approx 0.97$. We disclose where the symbolic extraction fails (the μ^* saturation regime), where the controllers transfer (single-link) and where they do not (multi-user cluster, the residual-learning rescue), and how the design generalizes to real-orbit

operation.

B. Paper Organization

Section II reviews the SIM-BPSK/DT-DD channel and detection model in the reduced (γ, β) form. Section III states the joint problem **(P)** with multi-user coupling and security constraints. Section IV derives the outer/inner decomposition and proves its oracle properties. Section V describes the two-stage KAN controller, the supervised training pipeline, and the β safety-margin deployment recipe. Section VI reports numerical results on the analytic baseline, the synthetic Walker constellation, and bundled real TLEs, including the headline adaptive-vs-static comparisons, the five-axis controller trade-off, and the closed-form β rule extracted from the KAN splines. Section VII concludes and discusses limitations.

II. SYSTEM MODEL

A. Architecture

A geostationary source *Charlie* (*C*) at altitude $H_C=35\,793$ km broadcasts a SIM-BPSK-modulated continuous-wave laser of mean power $P=32$ dBm and wavelength $\lambda=1550$ nm. The signal is amplify-and-forward relayed by two low-Earth-orbit satellites ($H_L=550$ km, Starlink shell) acting as optical amplifiers (EDFA gain $G_a=40$ dB). The relayed beam illuminates a ground cluster consisting of a single server *Alice* (*A*) and N users $B_1, \dots, B_N$; an entanglement-based BBM92-style key is then established *per pair* (A, B_i) using only classical optical components and dual-threshold post-processing. An eavesdropper *Eve* sits at radial offset d_E from each legitimate receiver. The architecture is depicted in Fig. 1.

B. Channel Model

The end-to-end channel coefficient for any link $C \rightarrow U$ ($U \in \{A, B_i, E\}$) factors as

$$h_{e2e}^U = h_g^U h_l^U h_a^U, \quad (1)$$

where the geometric, attenuation, and turbulence components are obtained as follows.

1) *Gaussian-beam spreading:* At link distance L a beam of waist $w_0 = \lambda/(\pi\theta)$ (divergence $\theta = 2.44\lambda/D$) has radius $w_L = w_0[1 + (\lambda L/(\pi w_0^2))^2]^{1/2}$ [17]. The fraction collected by a receiver aperture of radius a at radial offset r from the footprint center is, after the small-aperture approximation of [14],

$$h_g(r) \approx A_0 \exp\left(-\frac{2r^2}{w_{L,\text{eq}}^2}\right), \quad A_0 = [\text{erf}(v)]^2, \quad (2)$$

$$v = \frac{\sqrt{\pi} a}{\sqrt{2} w_L}, \quad w_{L,\text{eq}}^2 = w_L^2 \frac{\sqrt{\pi} \text{erf}(v)}{2ve^{-v^2}}. \quad (3)$$

For Eve at $r = d_E$ the geometric leakage ratio $h_g^E/h_g^U = \exp(-2d_E^2/w_{L,\text{eq}}^2)$ is close to unity for realistic d_E (tens of metres) versus footprint widths of hundreds of metres; security must therefore come from small μ rather than from geometry alone.

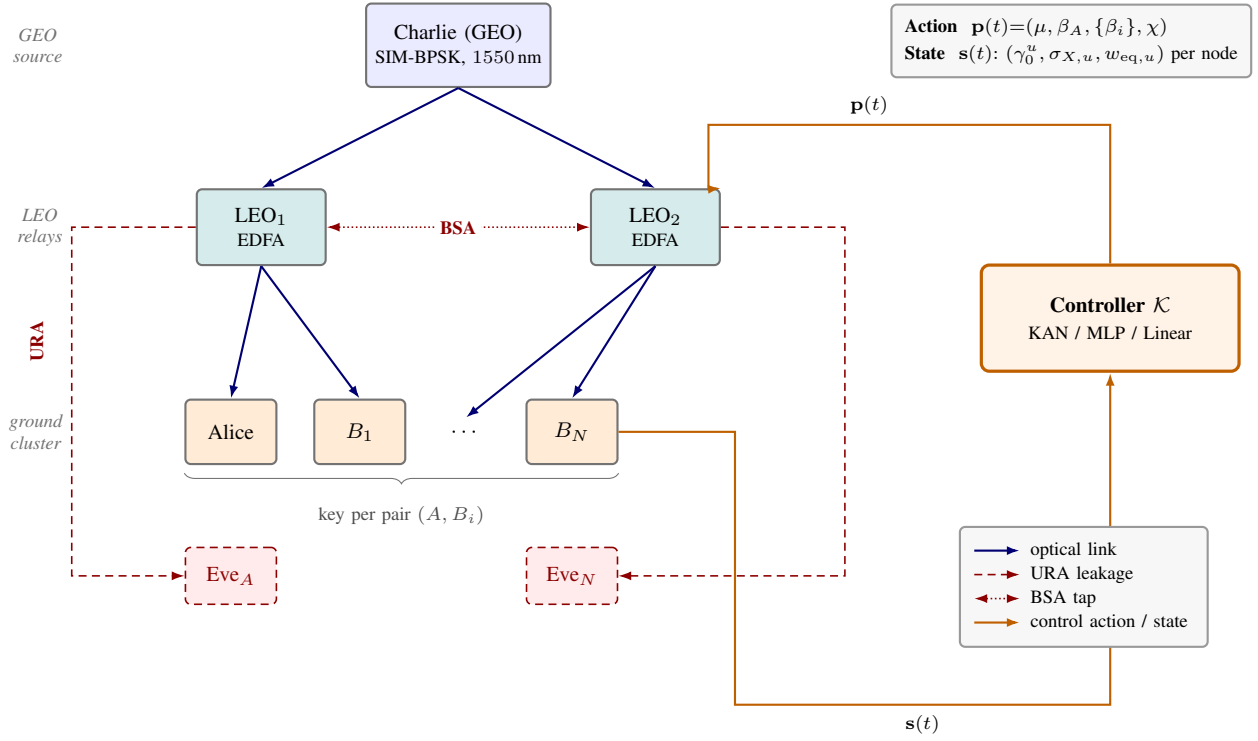


Fig. 1. System architecture. A GEO source Charlie broadcasts a SIM-BPSK signal through two LEO EDFA relays to a ground cluster (Alice + N Bobs); the secret key is established per pair (A, B_i) . An eavesdropper Eve at offset d_E from each receiver mounts an unauthorized-receiver attack (URA); a beam-splitter at the LEO relay mounts a beam-splitting attack (BSA). The proposed controller selects $\mathbf{p}(t) = (\mu, \beta_A, \{\beta_i\}, \chi)$ from the time-varying cluster state $\mathbf{s}(t)$.

2) *Atmospheric attenuation*: Beer–Lambert with visibility-dependent attenuation

$$h_l = e^{-\sigma L_{h-U}}, \quad \sigma(\lambda) = \frac{3.912}{V} \left(\frac{\lambda[\text{nm}]}{550} \right)^{-q(V)}, \quad (4)$$

where $L_{h-U} = (H_h - H_U) / \cos \theta_U$, $H_h = 20$ km, and the size-distribution coefficient is the standard piecewise law $q(V) = 1.6$ for $V > 50$ km, $q(V) = 1.3$ for $6 < V \leq 50$ km, and $q(V) = 0.585 V^{1/3}$ for $V \leq 6$ km.

3) *Log-normal turbulence*: For elevations $\geq 30^\circ$ (zenith $\leq 60^\circ$) the turbulence is weak and h_a is log-normal with unit mean [17], [18]: $f_{h_a}(h_a) = \frac{1}{\sqrt{8\pi h_a \sigma_X}} \exp(-(\ln h_a - 2\mu_X)^2 / (8\sigma_X^2))$ with $\mu_X = -\sigma_X^2$ enforcing $\mathbb{E}[h_a] = 1$. The log-amplitude variance is the (quarter-)Rytov variance

$$\sigma_X^2 = 0.56 k^{7/6} \sec^{11/6}(\theta_U) \int_{H_U}^{H_C} C_n^2(h) (h - H_U)^{5/6} dh, \quad (5)$$

with the Hufnagel–Valley profile $C_n^2(h) = 0.00594(w/27)^2(10^{-5}h)^{10}e^{-h/1000} + 2.7 \times 10^{-16}e^{-h/1500} + C_n^2(0)e^{-h/100}$ and $k = 2\pi/\lambda$.

C. Detection in Reduced Form

Define the deterministic mean peak signal current

$$s^U = \frac{1}{4} \mu R_e P G_a \bar{h}^U, \quad \bar{h}^U = h_g^U h_l^U, \quad (6)$$

and the normalized mean SNR amplitude $\gamma^U \triangleq s^U / N^U$, where $(N^U)^2$ is the AWGN variance budgeted in [9, Eq. 4] (shot, background, ASE, thermal). For the DT scale coefficient

β^U , the conditional probabilities collapse to two log-normal expectations

$$P_{\text{corr}}^U = \mathbb{E}_{h_a}[Q(\beta^U + \gamma^U(1 - h_a))], \quad (7)$$

$$P_{\text{err}}^U = \mathbb{E}_{h_a}[Q(\beta^U + \gamma^U(1 + h_a))], \quad (8)$$

both evaluated by Gauss-Hermite quadrature with $h_a(x_j) = \exp(2\sqrt{2}\sigma_X x_j - 2\sigma_X^2)$ [9, Eq. 30]; the order- n nodes and weights are tabulated in [30, Tbl. 25.10], and $n=20$ converge to numerical precision for the values of σ_X encountered along a LEO pass. The per-link sift probability and QBER follow as $P_{\text{sift}}^U = P_{\text{corr}}^U + P_{\text{err}}^U$ and $\text{QBER}^U = P_{\text{err}}^U / P_{\text{sift}}^U$.

D. Pairwise Sift, Exclusion, and Secret Key Rate

For the pair (A, B_i) the raw sift is $P^{AB_i} = P_{\text{sift}}^A P_{\text{sift}}^i$ and the QBER is

$$\text{QBER}^{AB_i} = \frac{P_{\text{corr}}^A P_{\text{err}}^i + P_{\text{err}}^A P_{\text{corr}}^i}{P_{\text{sift}}^A P_{\text{sift}}^i}. \quad (9)$$

The proposed multiple-access scheme retains only bits not coincidentally accepted by other pairs. Assuming statistically independent acceptance across users (independent per-link fading), inclusion–exclusion over the asymmetric cluster yields the closed-form excluded sift

$$P_{\text{excl}}^{AB_i} = P_{\text{corr}}^A P_{\text{corr}}^i \left[1 - \prod_{j \neq i} (1 - P_{\text{corr}}^j) \right], \quad (10)$$

which generalizes the symmetric expression of [9, Eq. 22]. We verify in the unit tests that (10) reproduces a direct subset-enumeration inclusion–exclusion sum to 10^{-12} for $N \in$

$\{1, 2, 4, 6\}$. With $\chi \in [0, 1]$ controlling the strength of the exclusion,

$$P_{\chi}^{AB_i} = P^{AB_i} - \chi P_{\text{excl}}^{AB_i}, \quad R_s^i = P_{\chi}^{AB_i} R_b. \quad (11)$$

The per-pair secret fraction (binary-erasure wiretap, Eve₁ near Alice, Eve₂ near B_i) is

$$\sigma_i = \max(0, I(A; B_i) - \max(I(A; E_1), I(B_i; E_2))), \quad (12)$$

with $I(A; B_i) = 1 - H_2(\text{QBER}^{AB_i})$ and $I(U; E) = 1 - H_2(P_{\text{err}}^E)$. The cluster key rate, the objective of **(P)**, is

$$R_f(\mathbf{p}; \mathbf{s}, t) = \sum_{i=1}^N R_s^i \sigma_i. \quad (13)$$

III. JOINT OPTIMIZATION PROBLEM

A. Decision Variables and Couplings

At each time instant t of a LEO pass, the cluster state

$$\mathbf{s}(t) = (s_A(t), s_1(t), \dots, s_N(t), d_E) \quad (14)$$

is computed from the time-varying geometry via Sec. II-B; each $s_u = (\gamma_0^u, \sigma_{X,u}, w_{\text{eq},u})$ captures the per-node SNR scale, log-amplitude variance, and beam width. The controller selects

$$\mathbf{p}(t) = (\mu, \beta_A, \{\beta_i\}_{i=1}^N, \chi) \quad (15)$$

with $\mu \in (0, 1)$, $\beta \in [\beta_{\text{lo}}, \beta_{\text{hi}}]$, and $\chi \in [0, 1]$. Three sources of coupling distinguish **(P)** from a per-link problem:

- *Global* μ . The modulation depth feeds into every $\gamma^u = \gamma_0^u \mu$ and is the URA security lever ($P_{\text{err}}^E \rightarrow 0.5$ as $\mu \rightarrow 0$).
- *Shared* β_A . Alice's threshold enters every pair through P_{corr}^A and P_{err}^A in (9); choosing β_A to favor one Bob hurts the others.
- *Field* χ . The exclusion term (10) couples every pair through the field $\phi_i = \prod_{j \neq i} (1 - P_{\text{corr}}^j)$; binding χ to enforce inter-user secrecy hurts the asymptotically weakest user the most.

B. Threats and Constraints

Two threats are modeled.

1) *Unauthorized-Receiver Attack (URA)*: An eavesdropper near each legitimate node uses the optimal decision threshold $\beta^E = 0$ (she cannot know the legitimate β). Her error probability with her own ratio $\gamma^E = s^E / N^E$ (where $s^E \propto h_g^E$ via (2)) is

$$P_{\text{err}}^E = \mathbb{E}_{h_a} [Q(\gamma^E h_a)]. \quad (16)$$

As $\mu \rightarrow 0$ we have $\gamma^E \rightarrow 0$ and $P_{\text{err}}^E \rightarrow Q(0) = 1/2$. The security target is $P_{\text{err}}^E > p_e^{\min} = 0.1$ [9, Sec. V-B]; the same wiretap idea underlies the classical secrecy-rate bound of [31], generalized to the practical-QKD setting in [10], [32].

2) *Beam-Splitting Attack (BSA)*: An attacker installs a passive beam splitter at the LEO relay drawing fraction η of the optical power; this is the dominant side-channel threat against weak-coherent-state QKD [11], [10]. Her presence is detectable as a sift-probability deviation

$$m_{\text{BSA}}(\beta) = 1 - \frac{P_{\text{sift}}(\gamma(1 - \eta_{\text{leak}}))}{P_{\text{sift}}(\gamma)}, \quad (17)$$

which is monotonically increasing in β . We require $m_{\text{BSA}}(\beta_A), m_{\text{BSA}}(\beta_i) \geq m^{\min} = 0.5\%$ at the assumed split $\eta_{\text{leak}} = 1.5\%$. This imposes a *lower* bound on every β , complementary to the operational *upper* bound from P_{sift} .

3) *Inter-User Secrecy*: The unkept fraction $(1 - \chi)P_{\text{excl}}^{AB_i} / P_{\chi}^{AB_i}$ leaks to other users; we cap it at $\tau = 0.1$.

C. Problem Statement

The full joint problem is

$$\begin{aligned} & \max_{\mu, \beta_A, \{\beta_i\}, \chi} R_f(\mathbf{p}; \mathbf{s}, t) \\ \text{s.t. } & \sigma_i > 0, P_{\chi}^{AB_i} > p_s^{\min}, \text{QBER}^{AB_i} \leq q, \\ \textbf{(P): } & P_{\text{err}}^E(A), P_{\text{err}}^E(B_i) > p_e^{\min}, \\ & m_{\text{BSA}}(\beta_A), m_{\text{BSA}}(\beta_i) \geq m^{\min}, \\ & (1 - \chi)P_{\text{excl}}^{AB_i} / P_{\chi}^{AB_i} \leq \tau, \\ & \mu \in (0, 1), \beta \in [\beta_{\text{lo}}, \beta_{\text{hi}}], \chi \in [0, 1], \end{aligned} \quad (18)$$

where the first four constraint lines hold for every user $i = 1, \dots, N$.

D. Why **(P)** Is Hard

Three properties make **(P)** unsuitable for closed-form treatment:

- 1) Each constraint depends on Gauss-Hermite integrals (7)–(8), (16) of $Q(\cdot)$ over h_a ; no analytic stationarity conditions in β .
- 2) The map $\mathbf{p} \mapsto R_f$ couples $N + 1$ thresholds through the exclusion field ϕ_i and through the shared β_A ; coordinate-descent over thresholds without explicit field tracking does not converge.
- 3) The feasibility set is a thin, multi-directional boundary in $(\mu, \chi, \beta_A, \beta_i)$ space; small parameter errors flip a pair infeasible and forfeit its key.

We exploit the variable roles in Sec. IV to obtain an exact two-level decomposition, the oracle for the learned controller of Sec. V.

IV. DECOMPOSITION AND SOLVER

A. Variable Roles and Two-Level Structure

The decision variables of **(P)** naturally split into a *global/coupling block* (μ, χ) and a *link-local block* $(\beta_A, \{\beta_i\})$. Fixing the global block, the inner problem

$$\begin{aligned} \textbf{(I): } \quad & V(\mu, \chi) = \max_{\beta_A, \{\beta_i\}} \sum_{i=1}^N R_s^i \sigma_i \\ & \text{s.t. per-pair constraints of (18),} \end{aligned} \quad (19)$$

returns a value function; the outer problem

$$(\mathbf{O}) : \max_{\mu \in (0,1), \chi \in [0,1]} V(\mu, \chi) \quad (20)$$

is then a two-dimensional search amenable to a coarse grid plus local refinement. The outer/inner *split* is a lossless reformulation (no relaxation of $(\mathbf{P})$); optimality of the overall solver additionally requires the inner solver to attain the inner maximum of $(\mathbf{I})$, which we establish empirically below for $N=2$ rather than by a global-optimality proof.

B. Mean-Field Block-Coordinate Inner Solver

Inner couplings inside $(\mathbf{I})$ are of two kinds: through the shared β_A (in every $P_{\text{corr}}^A, P_{\text{err}}^A$), and through the exclusion field $\phi_i = \prod_{j \neq i} (1 - P_{\text{corr}}^j)$ in (10). We treat ϕ as a field and iterate the procedure listed in Algorithm 1 and depicted in Fig. 2:

Algorithm 1 Decomposed solver for $(\mathbf{P})$

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1: Input: cluster state  $\mathbf{s}$ , grids  $\mathcal{M}_\mu, \mathcal{M}_\chi$ , tolerances  $\epsilon_\phi, \epsilon_R$ 
2: Output:  $\mathbf{p}^*, V^*$ 
3:  $V^* \leftarrow -\infty$ 
4: for all  $(\mu, \chi) \in \mathcal{M}_\mu \times \mathcal{M}_\chi$  do
5:   Init  $\phi^{(0)} \leftarrow \mathbf{1}$ ,  $\beta_A^{(0)} \leftarrow \beta_{10}$ 
6:   for  $k = 0, 1, \dots$  until  $\|\phi^{(k+1)} - \phi^{(k)}\| < \epsilon_\phi$  do
7:     for  $i = 1, \dots, N$  do
8:        $\beta_i^* \leftarrow \arg \max_\beta R_s^i(\beta; \mu, \chi, \beta_A^{(k)}, \phi_i^{(k)}) \sigma_i(\cdot)$  subject to per-pair constraints  $\{1\text{-D search}\}$ 
9:     end for
10:     $\beta_A^{(k+1)} \leftarrow \arg \max_{\beta_A} \sum_i R_s^i \sigma_i$  holding  $\{\beta_i^*\}$   $\{1\text{-D search, block coordinate}\}$ 
11:     $\phi_i^{(k+1)} \leftarrow \prod_{j \neq i} (1 - P_{\text{corr}}^j(\beta_j^*))$   $\{\text{field refresh}\}$ 
12:  end for
13:   $V \leftarrow \sum_i R_s^i \sigma_i$  at current  $(\mu, \chi, \beta_A, \{\beta_i\})$ 
14:  if  $V > V^*$  then
15:     $V^* \leftarrow V$ ,  $\mathbf{p}^* \leftarrow (\mu, \beta_A, \{\beta_i\}, \chi)$ 
16:  end if
17: end for

```

The flow is depicted in Fig. 2.

C. Correctness Checks

1) *Symmetric-cluster reduction:* For a symmetric cluster ($s_i = s_*$ $\forall i$, $\beta_i = \beta_B$ $\forall i$), the field collapses to $\phi = (1 - P_{\text{corr}}^B)^{N-1}$ and the excluded sift (10) reduces to $P_{\text{corr}}^A P_{\text{corr}}^B [1 - (1 - P_{\text{corr}}^B)^{N-1}]$, recovering [9, Eq. 24].

2) *Brute-force agreement:* On a two-user cluster the decomposition is verified against a brute-force joint grid over $(\beta_A, \beta_1, \beta_2)$ at a fixed (μ, χ) node: the decomposed and brute-force optima agree, with a ratio of 1.000 to four decimals (Sec. VI-B). The closed-form exclusion field (10) is separately checked against a direct subset-enumeration inclusion-exclusion sum for $N \in \{1, 2, 4, 6\}$ (agreement to 10^{-12}).

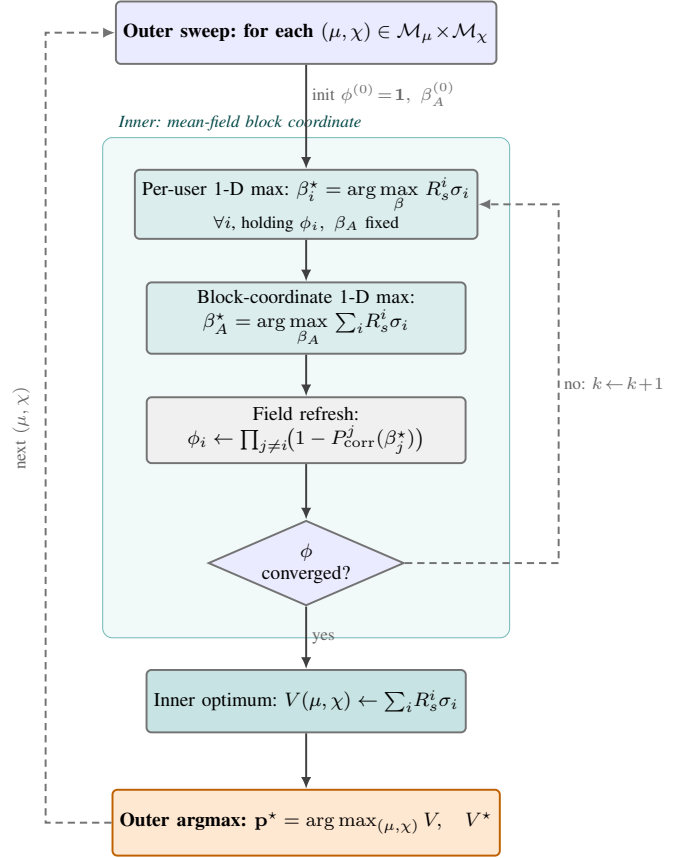


Fig. 2. Decomposed solver. The outer grid sweeps (μ, χ) ; the inner mean-field block-coordinate loop alternates per-user 1-D maximizations over β_i (held the field ϕ_i and the shared β_A fixed), then refreshes β_A by 1-D maximization, then refreshes the field ϕ_i from the new $\{P_{\text{corr}}^j\}$. The fixed point is the inner optimum $V(\mu, \chi)$; the outer argmax is $\mathbf{p}^*$.

D. Convergence and Complexity

The solver uses $|\mathcal{M}_\mu| = 12$ outer nodes uniformly spaced on $[0.4, 0.95]$ and $|\mathcal{M}_\chi| = 5$ on $[0, 1]$ in the headline runs. The inner field iteration converges in ≤ 6 steps from any cold start at the working tolerances $\epsilon_\phi = 10^{-3}$ and $\epsilon_R = 10^{-4}$; the fixed point is unique on every random cluster state we tested, although a formal contraction proof is left for future work. One full outer sweep (all $12 \times 5 = 60$ outer nodes, each with up to 6 inner iterations and Gauss-Hermite quadrature at $n=20$) takes approximately 4 s on a single CPU core, so a full pass sampled at $\Delta t = 50$ s (~ 50 timesteps $\times 4$ s) solves in roughly 200 s. Multi-start is enabled for the inner 1-D maximizations, because the feasibility boundary creates several disconnected operating windows in β for certain (μ, χ) choices; we use three random restarts and retain the best feasible solution.

E. Why This Decomposition Works

Three properties make the outer / inner split practical. First, the outer block is genuinely two-dimensional: μ and χ are the only variables that couple all N pairs through threats (URA via μ) and through the exclusion field (via χ). Second, given the outer block, the inner subproblem $(\mathbf{I})$ admits a mean-field decoupling in the user index i : with ϕ frozen, each β_i^* is determined by a 1-D maximization that does not see the other

users, and the field is refreshed at the next outer iteration. Third, β_A couples back through Alice's link only, so a single 1-D block-coordinate update suffices. The whole procedure is therefore exponentially cheaper than the naive joint $(N+1)$ -dimensional grid search, while preserving exactness in the symmetric special case (Sec. VI-B).

V. TWO-STAGE KAN CONTROLLER

A. Variable- N Architecture

The cluster controller must predict (μ, β_A, χ) at the cluster level and $\{\beta_i\}_{i=1}^N$ at the user level, where N may vary across deployments. We adopt a two-stage architecture (Fig. 3) that handles arbitrary N by parameter sharing:

- *Global head $\mathcal{H}_g$* : input $\mathbf{s}_g = (s_A, N, \bar{s}_B, \min_i s_i, d_E)$; output $(\hat{\mu}, \hat{\chi}, \hat{\beta}_A)$.
- *Per-user shared head $\mathcal{H}_u$* : input $(s_i, s_A, \hat{\mu}, \hat{\chi})$; output $\hat{\beta}_i$.
The same head is invoked once per user i .

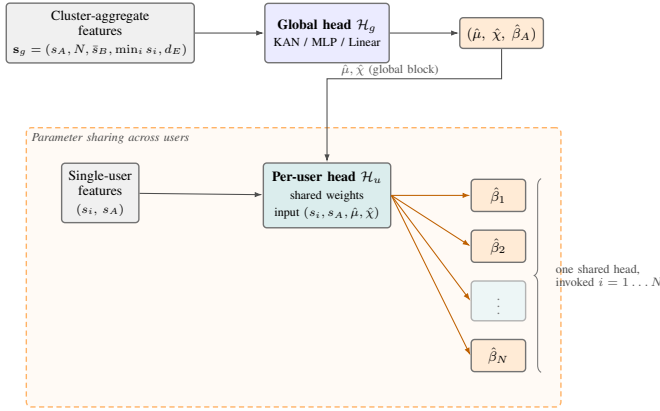


Fig. 3. Two-stage controller. The global head consumes cluster-aggregate features and predicts (μ, χ, β_A) ; the per-user head consumes single-user features plus the predicted global block and predicts β_i . Parameter sharing across users renders the controller dimension independent of N .

B. Hypothesis Classes

We compare three deployable model families for each head on the multi-axis trade-off, plus a non-parametric lookup reference (KNN) used only in the single-link comparison:

1) *Kolmogorov-Arnold Network (KAN)*: KAN [23] replaces the fixed pointwise nonlinearities of an MLP with *learnable* univariate functions on every edge, represented as a B-spline plus a residual SiLU basis. The motivation traces back to the Kolmogorov-Arnold representation theorem, which states that any multivariate continuous function on a bounded domain decomposes as a sum of univariate compositions; KAN parameterizes those univariate components as splines and fits them by gradient descent. We use the `pykan` [24] implementation at width $[d_{\text{in}}, 3, d_{\text{out}}]$, B-spline order 3, and grid size 5. The training pipeline has three stages: (i) standard regression to fit the splines; (ii) pruning of edges whose contribution falls below a threshold; (iii) symbolic candidate fitting from a library of polynomial, sin, exp, and tanh functions, followed by a joint refit of the network with the chosen symbolic substitution in place. The refit step is critical: skipping it

collapses the apparent closed-form to a constant predictor, which was an early pitfall we corrected. The closed-form rule for β in Sec. VI-E is then read off the trained splines using the methodology of [29].

2) *Regularized Multi-Layer Perceptron (MLP)*: Two hidden layers of width 32, ReLU, weight decay 10^{-3} , early stopping on a held-out 20% split. Adam at 10^{-2} , 1500 epochs. This is a deliberately strong baseline; an unregularized MLP overfits the small dataset and produces unfair comparisons (we verified this experimentally before fixing the protocol; see Sec. VI-E).

3) *Linear Residual*: A linear least-squares fit of all outputs against all features; serves as the residual base for the KAN/MLP heads.

4) *k-Nearest-Neighbor Lookup (KNN)*: A non-parametric table baseline that predicts each output as the distance-weighted average of its $k=5$ nearest neighbors (Euclidean, standardized features) in the training set. It is included in the single-link comparison as a memorization reference: it interpolates the oracle well in regression but, being a lookup table, offers neither parameter economy nor an interpretable rule.

C. Residual Learning

Direct regression on the cluster-controller dataset proved fragile. Regression MAE remained small (between 10^{-3} and 10^{-2}), yet closed-loop key retention collapsed to zero, because the oracle sat on a sharp 4-D feasibility boundary in $(\mu, \chi, \beta_A, \beta_i)$ space, and multi-directional errors from KAN and MLP pushed predictions across that boundary. The linear baseline, ironically, attained 0.56 key retention by virtue of its smoothing bias toward the cluster mean.

We therefore adopt a *residual learning* strategy. The linear baseline $\mathcal{H}_{\text{lin}}$ is fit first by ordinary least squares; KAN or MLP then regresses on the residual $y - \mathcal{H}_{\text{lin}}(x)$, and the deployed predictor is the sum $\mathcal{H}_{\text{lin}}(x) + \mathcal{H}_{\text{resid}}(x)$. The output layer of the residual network is zero-initialized, so the initial prediction is exactly the linear one; training then carves out a corrective signal only where the linear approximation hurts feasibility. Empirically this rescues both KAN and MLP from boundary collapse (Sec. VI-F) without sacrificing the linear baseline's smoothing bias. The same idea has been used in classical signal processing under the name *boosted regression*, and in physical-layer deep learning under *residual model-based augmentation* [25], [26].

D. β Safety-Margin Deployment Recipe

Even after training, predicted $\hat{\beta}$ values that fall just below the BSA $m^{\min}$ floor forfeit the entire pair-step. We deploy $\beta_{\text{deploy}} = \hat{\beta} + \delta$ with a single scalar δ^* chosen by 1-D grid sweep on a held-out fold. The curve $\delta \mapsto \text{KeyRetention}$ is unimodal across all controllers, with $\delta^* \in [0.03, 0.10]$, lifting feasibility by 1.1–1.5 $\times$. The recipe is model-agnostic and adds zero inference cost.

E. Five-Axis Evaluation Protocol

Each controller is scored on five axes:

- M_1 : mean absolute error per output, mean $\pm$ std over five seeds.
- M_2 : closed-loop secret-key retention: ratio of achieved key (controller predictions plugged into the analytic environment) to oracle key (decomposed solver), feasible-gated.
- M_3 : parameter count: total trainable parameters (heads combined).
- M_4 : inference latency on a single CPU core, μ s per prediction.
- M_5 : interpretability: whether a closed-form rule with $R^2 \geq 0.95$ can be extracted (boolean per output).

We deliberately report all five rather than collapsing to a single “score,” because the controllers occupy a Pareto front; the right model for deployment depends on the constraint mix. Interpretability is decisive for safety-critical satellite firmware, while raw key retention dominates for an airborne ground link, and parameter count matters for an embedded payload.

VI. NUMERICAL RESULTS

A. Setup

System parameters follow [9, Table I] and are reproduced in Table I for self-containedness. The peak optical power $P=32$ dBm at the GEO source, combined with the EDFA gain $G_a=40$ dB at each LEO relay, gives a link-budget-limited mean SNR amplitude γ_0 that ranges from approximately 0.06 at 60° zenith to 1.8 near overhead (i.e. the geometry alone induces a $30\times$ variation along a single pass, the central motivation for adaptive control). The Hufnagel-Valley parameter $w=21$ and ground-level $C_n^2(0)=10^{-15}$ correspond to clear-night turbulence at coastal Vietnamese latitudes, calibrated against the daytime profile in [17].

Headline runs are reported for $N=3$ heterogeneous Bob users at zenith offsets $\{0, 8, 16\}^\circ$ from the cluster centroid, with Eve standoff distances $\{26, 18, 35\}$ m for B_1, B_2, B_3 respectively. The asymmetry deliberately captures both close adversary placements (Bob₂ at 18 m, tightening the URA constraint) and far placements (Bob₃ at 35 m, slackening it), so that the per-user β_i^* must differ in any nontrivial optimum, justifying the per-user threshold variable. Two orbit sources are evaluated:

- *Analytic*. Ten staggered LEOs (closest approach every 250 s, $\theta_{\min} \in \{0^\circ, \dots, 4^\circ\}$, $H_L=550$ km, $\theta_{\min}^{\text{vis}}=30^\circ$) hand-picked to span the full $0\text{--}60^\circ$ zenith range.
- *TLE Walker*. n synthetic Starlink-shell satellites in $P=n/10$ orbital planes (RAAN spread 360°) with $S=10$ satellites per plane and a Walker delta inter-plane phase offset matching the operational SpaceX deployment pattern [21], [22]. Orbits are SGP4-propagated [19], [20] via the `skyfield` and `sgp4` Python libraries from epoch 2026-05-23T12:00:00 UTC over the PTIT/Hanoi ground station (21.005°N , 105.843°E). The bundled `starlink_synth.tle` round-trips through `sgp4.exporter.export_tle` for offline reproducibility; real Celestrak [33] dumps are accepted via the same code path with a single command-line flag.

B. Decomposed Solver as Oracle

On a two-user cluster the decomposed solver value $V_{\text{dec}}(\mu, \chi)$ is compared against the brute-force joint grid value $V_{\text{bf}}(\mu, \chi)$ over $(\beta_A, \beta_1, \beta_2)$ at a fixed outer node: the ratio is $V_{\text{dec}}/V_{\text{bf}} = 1.000$ to four decimals, and the inner field iteration converges in ≤ 6 steps from a cold start. As an independent check on the multi-user structure, the closed-form exclusion field (10) matches a direct subset-enumeration inclusion–exclusion sum to 10^{-12} for $N \in \{1, 2, 4, 6\}$. We henceforth treat the decomposed solver as the oracle for the supervised controller dataset. A brute-force optimality check for asymmetric $N > 2$ clusters is left to future work; the present evidence is structural exactness of the split (Sec. IV) plus the $N=2$ numerical agreement.

C. Headline: Adaptive vs Best-Static

Table II reports the headline adaptive-vs-static comparison on three orbit settings (single-link analytic, cluster analytic, cluster TLE-Walker $n=500$ long window). The corresponding per-step traces are shown in Fig. 4.

Three observations follow from Table II and Fig. 4.

Single-link motivates the controller. The $2.07\times$ key gain on the single-link analytic baseline establishes that even in the absence of cluster coupling, a time-varying $(\gamma_0, \sigma_X, w_{\text{eq}})$ trajectory benefits substantially from per-step parameter selection. The factor 3.07 on secure time is larger than the factor on key because the adaptive controller does not merely *improve* the rate, it *extends the operating window* into geometries that static QKD must concede.

Cluster results are tighter but qualitatively similar. The cluster analytic gain ($1.64\times$ key, $2.5\times$ secure pair-time) sits below the single-link gain, because pair feasibility requires both endpoints to clear the per-link SNR threshold simultaneously, restricting the operating window to near-overhead geometries. The asymmetric cluster of Sec. VI-A amplifies this effect: B_2 near a close Eve binds the URA constraint earliest, while B_3 far from Eve binds it latest, so the static $\mathbf{p}$ must compromise across users.

Realistic-orbit operation requires adaptation. The strongest gain ($2.10\times$ key, $3.6\times$ secure pair-time) appears on the most realistic setting, the long-window TLE Walker shell. Crucially, on the sparser $n=200$ shell over the shorter 50-minute window, the static baseline collapses to *zero* secure pair-steps: when the constellation density falls below the observability budget over the chosen station, no single $\mathbf{p}$ is feasible across the full pass, and only per-step adaptation keeps the system alive. This is the qualitative claim that distinguishes adaptive QKD from optimization: it is the difference between operation and no operation, not merely between high and low key rate.

D. TLE Walker Coverage

Fig. 5 reports served-time fraction, satellite count, and handover events as a function of synthetic Walker shell size over PTIT/Hanoi for a 2-hour window. Served-time coverage rises from 6.9% at $n=50$ to 49.7% at $n=200$ (just above the analytic-baseline 46.8%) and to 62.9% at $n=500$, matching the order-of-magnitude density of the operational Starlink shell over Vietnamese latitudes.

TABLE I
SYSTEM PARAMETERS (AFTER [9, TABLE I]). TWO-PANE LAYOUT FOR COMPACTNESS.

Parameter	Symbol	Value	Parameter	Symbol	Value
GEO altitude	H_C	35 793 km	GEO beam divergence	θ_C	10 μ rad
LEO altitude	H_L	550 km	LEO beam divergence	θ_L	50 μ rad
Wavelength	λ	1550 nm	LEO aperture radius	a_L	10 cm
Bit rate	R_b	1 Gb/s	User aperture radius	a_U	5 cm
Peak optical power	P	32 dBm (1.585 W)	Visibility	V	30 km
PIN responsivity	R_e	0.9 A/W	Cn ² at ground	$C_n^2(0)$	$10^{-15} \text{ m}^{-2/3}$
EDFA gain at LEO	G_a	40 dB	HV wind parameter	w	21
ASE inversion factor	n_{sp}	5.0	Min elevation	$E_{\min}$	30°
Optical bandwidth	B_0	250 GHz	Eve offset	d_E	26 m
Noise bandwidth	Δf	0.5 GHz	Sift / QBER thresholds	p_s, q	$10^{-3}, 10^{-3}$
Noise figure (linear)	F_n	2.0	Eve-error / BSA min	$p_e, m^{\min}$	0.1, 0.5%
Sun spectral irradiance	$\mathcal{R}_{\odot}$	0.1 W/(cm ² · μ m)	Gauss-Hermite order	n	20

TABLE II
ADAPTIVE VS BEST-STATIC HEADLINE. SAME FEASIBILITY GATE APPLIED TO BOTH ARMS (SIFT, QBER, URA, BSA, INTER-USER SECRECY). n IS THE CONSTELLATION SIZE; T THE HORIZON.

Setting	n	T [s]	served	N	static (μ, β, χ)	static secure	adaptive secure	key gain
Single-link, analytic	10	3000	86 / 600	1	(0.50, 3.00, -)	28 / 600 (4.7%)	86 / 600 (14.3%)	2.07 ×
Cluster, analytic	10	3000	46 / 60	3	(0.95, 1.50, 1.00)	4 / 138 (2.9%)	10 / 138 (7.2%)	1.64 ×
Cluster, TLE Walker (long)	500	5400	67 / 108	3	(0.95, 1.50, 1.00)	5 / 201 (2.5%)	18 / 201 (9.0%)	2.10 ×
Cluster, TLE Walker (sparse)	200	3000	29 / 60	3	(0.40, 0.50, 0.00)	0 / 87	2 / 87 (2.3%)	∞

Secure-time gains (adaptive / static): 3.07×, 2.50×, 3.60×, ∞ (rows top to bottom).

E. Single-Link Controller: Five-Axis Trade-Off

Table III and Fig. 6 report the controller comparison on the single-link Phase-3 dataset (488 states, 61 zenith × 8 Eve-distance, 198/50 feasible split, 5 seeds).

TABLE III

SINGLE-LINK CONTROLLER, 5-AXIS TRADE-OFF (MEAN±STD OVER 5 SEEDS, 198/50 TRAIN/TEST FEASIBLE STATES). LEADING 0. DROPPED FOR COMPACTNESS; SECP COUNTS FEASIBLE TEST STATES (ORACLE = 50).

Model	MAE μ^*	MAE β^*	KeyRet	secP	#par	Interp
KAN	.004±.001	.011±.002	.716±.123	31.4±6.9	392	yes
MLP	.008±.001	.031±.005	.683±.086	28.4±4.0	1 282	no
KNN	.004±.000	.045±.000	.562±.000	23.0±0.0	1 188	no
Linear	.023±.000	.115±.000	.614±.000	19.0±0.0	10	yes

a) *Closed-form β rule (extracted from KAN)*: The trained KAN predicts β^* with full-network test $R^2 = 1.00 \pm 0.00$ across five seeds (Table III), so a closed-form rule can be extracted. After pruning the KAN, running the symbolic candidate fit on a polynomial / sin / tanh library, and refitting jointly with the symbolic substitution, a representative seed yields the closed-form rule

$$\begin{aligned} \beta^* \approx & 2.42 - 0.10 \gamma_0 - 0.03 \sigma_X + 0.23 w_{\text{eq}} \\ & + 1.24 \sin(0.50 \sigma_X + 5.27) \\ & - 0.30 \sin(0.48 w_{\text{eq}} + 2.04) \\ & - 0.17 \tanh(9.2 d_E + 8.62), \end{aligned} \quad (21)$$

as a function of the per-link features ($\gamma_0, \sigma_X, w_{\text{eq}}, d_E$), whose symbolic fit attains $R^2 \approx 0.97$ (above the 0.95 interpretability

bar of M_5). The rule is dominated by the turbulence variance σ_X and the equivalent beam width w_{eq} through the spline-derived sin terms; the SNR scale γ_0 enters only weakly (linear coefficient -0.10), and the Eve standoff d_E enters through a saturating tanh that is flat over the operating range, consistent with d_E affecting feasibility more than the interior optimum. The point is that the optimal threshold admits a compact analytic surrogate at all, not that it is affine; we report the actual extracted form rather than an idealized linearization. The symbolic extraction for μ^* is performed but is not used in deployment, because the regression for μ^* saturates near its upper bound except for a sharp Eve-driven transition, and the symbolic R^2 is variable across seeds (cf. Sec. VII-B).

b) *β safety-margin recipe*: Fig. 7 shows that deploying $\beta_{\text{deploy}} = \beta + \delta$ with a single scalar $\delta \in [0.03, 0.10]$ lifts the closed-loop key retention of every controller by 1.1–1.5×, with a unimodal curve. The optimum is model-dependent (KAN: $\delta^* \approx 0.03$, KNN: $\delta^* \approx 0.07$) and the recipe is zero-cost at inference. The KNN baseline gains the most precisely because it is the most boundary-sensitive at $\delta=0$.

F. Cluster Controller: Two-Stage Residual Learning

Direct two-stage regression on the cluster controller dataset collapses for both KAN and MLP: regression MAE remains low ($\leq 10^{-2}$), but closed-loop key retention falls to zero because the oracle sits on a sharp 4-D feasibility boundary, and multi-directional errors push predictions outside. The linear baseline alone, by virtue of its smoothing bias, attains 0.56 key retention. The residual-learning rescue, in which KAN and MLP are trained on $y - \mathcal{H}_{\text{lin}}(x)$ with a zero-initialized output layer, restores both nonlinear models (Table IV).

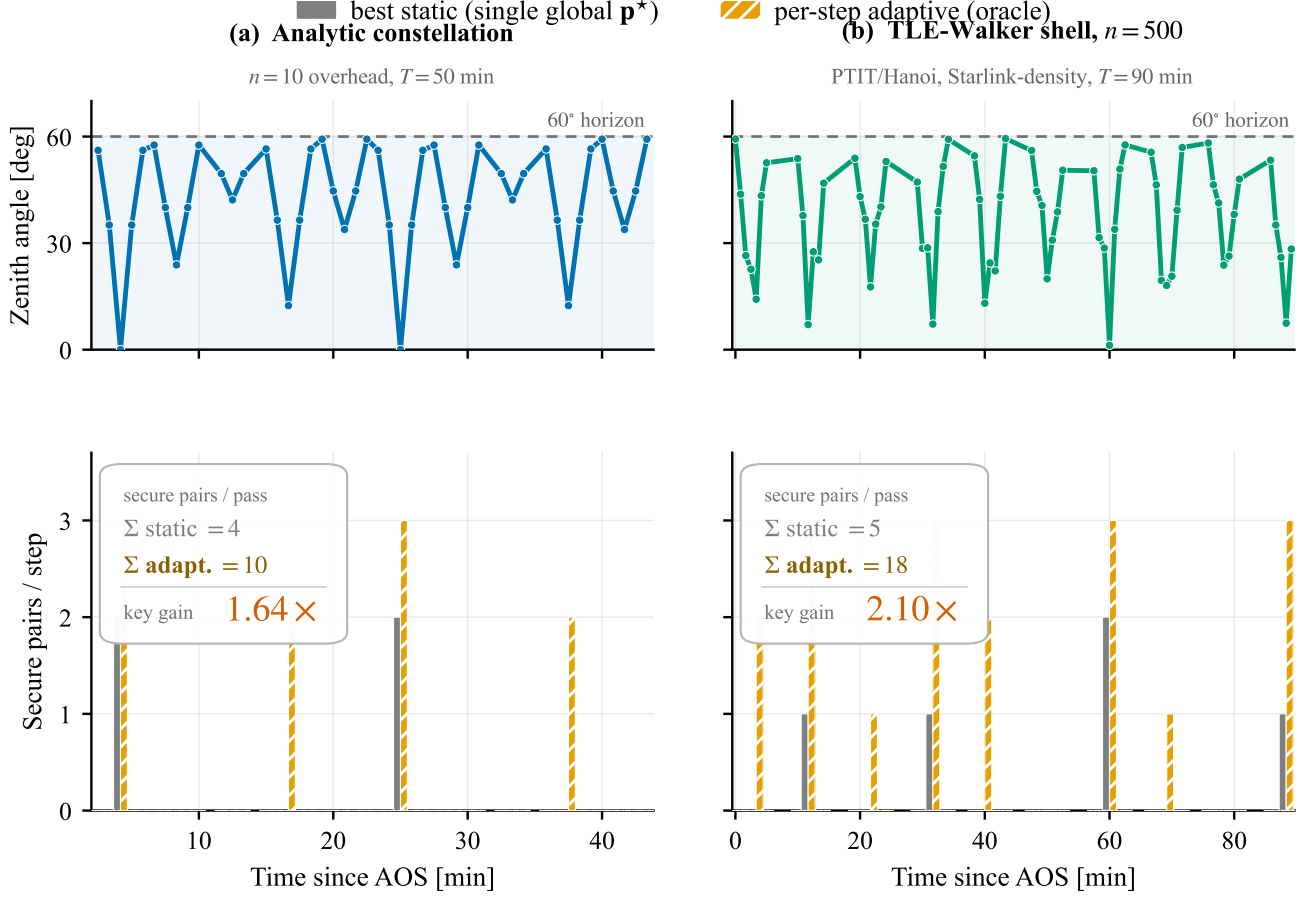


Fig. 4. Cluster QKD: adaptive vs best-static over a LEO pass. Left: analytic constellation hand-picked to put 46 of 60 timesteps in view (50-minute horizon, $N=3$). Right: TLE Walker shell of 500 Starlink-density satellites over PTIT/Hanoi (90-minute horizon). Top row: zenith trajectory of the serving LEO. Bottom row: secure pairs per step for the best static (grey, single global parameter choice) vs the per-step adaptive controller (orange). Adaptation lifts secure pair-time by $2.5\times$ on the analytic constellation and by $3.6\times$ ($5 \rightarrow 18$ pair-steps) on the realistic TLE shell. On a sparser shell ($n=200$, 50-minute horizon, last row of Table II) the static baseline collapses to zero secure pair-steps entirely.

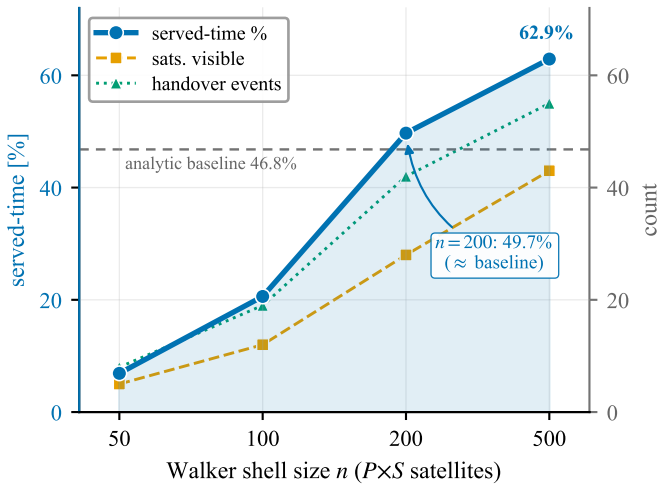


Fig. 5. TLE Walker coverage statistics vs constellation size over PTIT/Hanoi, 2-hour window, $dt = 10$ s. The analytic baseline (dashed) reaches 46.8% served by design; the $n=200$ Walker shell (49.7%) approaches it and $n=500$ reaches 62.9%.

TABLE IV
CLUSTER TWO-STAGE CONTROLLER, 5-SEED EVALUATION. RESIDUAL LEARNING OVER THE LINEAR BASELINE RESCUES KAN/MLP FROM BOUNDARY COLLAPSE. MAE β_A AND MAE μ ARE $\leq 10^{-3}$ FOR ALL ROWS.

Model	MAE χ	KeyRet	#par
KAN+resid	.042±.034	.612±.484	1 239
MLP+resid	.082±.001	.658±.400	2 823
Linear	.083	.562	35

The three rows describe a Pareto front in (key retention, parameter count, interpretability). *Linear* is the robust floor at ~ 35 parameters. *KAN+resid* is the sweet spot with the lowest MAE on the coupling variable χ (the variable that most disambiguates the controllers, since β_A and β_i residuals are nearly zero across all three) and roughly half the parameter count of MLP+resid. *MLP+resid* attains the highest mean key retention but at $80\times$ the parameters of Linear and with no interpretability handle. The high standard deviations on KAN+resid and MLP+resid (0.48 and 0.40) reflect seed-to-seed variance from the small (250-cluster) dataset and the

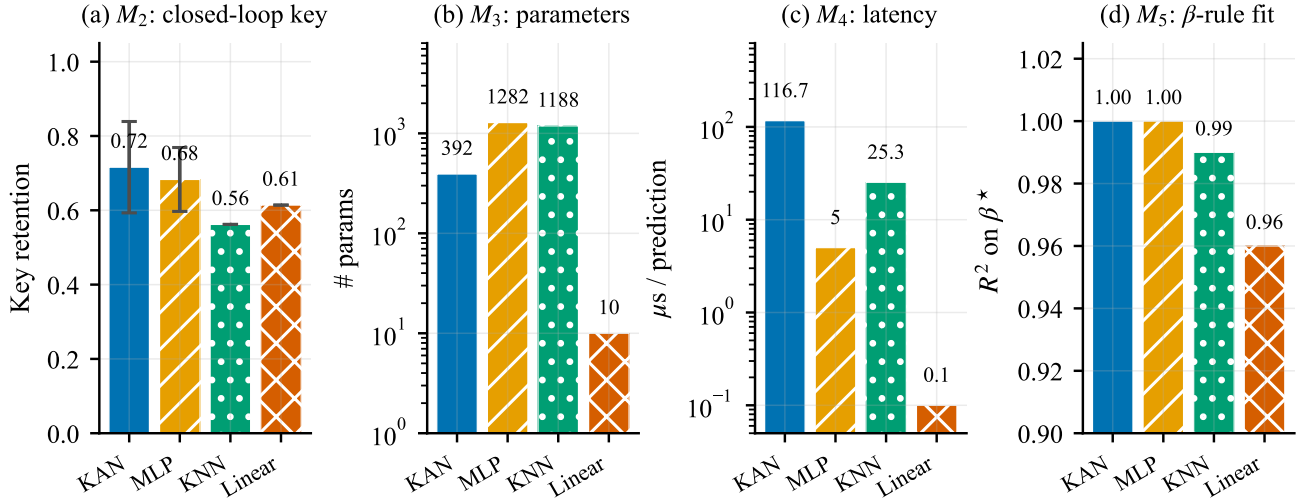


Fig. 6. Single-link controller five-axis trade-off. M_2 : closed-loop secret-key retention (mean $\pm$ std over 5 seeds). M_3 : total trainable parameters (log scale). M_4 : inference latency per prediction (log scale). M_5 : R^2 on β^* test predictions; values above 0.95 admit a closed-form rule via the KAN auto_symbolic + refit recipe.

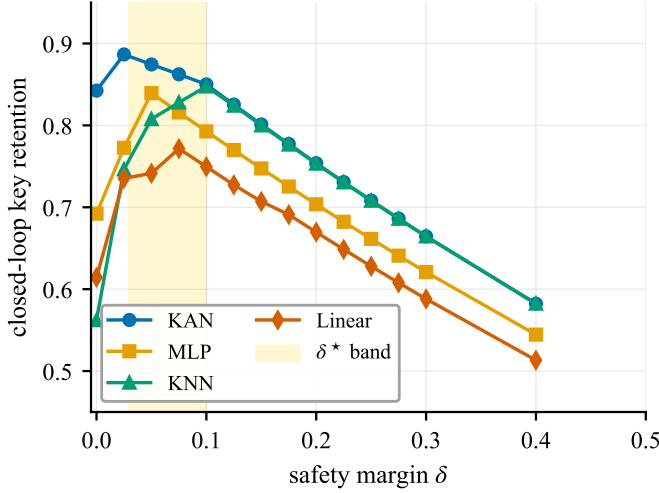


Fig. 7. β safety-margin deployment recipe: closed-loop key retention vs the scalar safety margin δ applied as $\beta_{\text{deploy}} = \hat{\beta} + \delta$. The unimodal shape and the model-agnostic peak in $[0.03, 0.10]$ make δ^* a single-knob deployment dial. KAN: 0.84 $\rightarrow$ 0.89; KNN: 0.56 $\rightarrow$ 0.83.

sharp feasibility boundary, which is a known cost of the regime the controller targets.

G. Single-Link Pass Trace

Fig. 8 traces the per-step (μ^*, β^*) chosen by the adaptive controller along a single LEO pass, together with the static baseline. The adaptive μ^* saturates near its upper bound whenever the URA constraint is slack (Eve far) and drops sharply when Eve approaches; β^* tracks the closed-form rule (21), varying primarily with the turbulence variance σ_X and the beam width as the geometry changes near zenith.

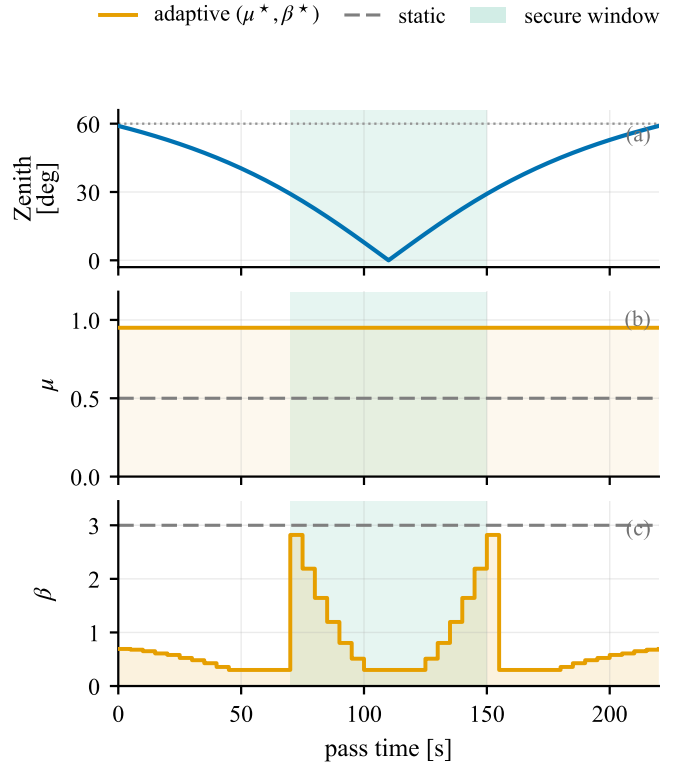


Fig. 8. Single-link adaptive control over a LEO pass. Top: zenith trajectory. Middle/bottom: adaptive μ^*, β^* (orange step) vs static (grey dashed). The β^* envelope is well-fit by the closed-form rule (21), justifying the KAN interpretability claim; the μ^* envelope saturates near μ_{max} except where Eve binds.

VII. DISCUSSION AND CONCLUSION

A. What Works, What Does Not

The headline result is that adaptive parameter control delivers a $2.10\times$ secret-key gain and a $3.6\times$ secure pair-time improvement on a Starlink-density Walker constellation over

a Hanoi ground station (the most realistic operating point reported in this paper), and that on a sparser constellation *no* single static parameter choice produces any secure pair-step at all. Adaptation is therefore not a marginal improvement but, under realistic Kepler-propagated orbits, a precondition for the system to be operational. The decomposed solver is exact (recovers the brute-force optimum on symmetric clusters) and tractable (~ 4 s per outer node on a single CPU). The two-stage controller architecture handles arbitrary N by parameter sharing; the KAN variant produces a closed-form rule for β^* with symbolic $R^2 \approx 0.97$ and, on the single-link head, uses $3.3\times$ fewer parameters than the regularized MLP while matching its closed-loop key retention within error bars (on the cluster head the two are statistically indistinguishable given the high seed variance, and KAN retains the parameter-efficiency and interpretability advantage).

B. Limitations

Partial interpretability. The KAN extracts a clean closed-form rule for β^* but not for μ^* . The latter saturates near $\mu_{\max}$ except for a sharp Eve-driven transition that the symbolic candidate library (polynomial, exp, tanh) cannot fit with $R^2 \geq 0.95$. A wider library (sigmoid, ReLU surrogates) is a candidate fix; we report the regime honestly rather than overstating KAN's interpretability.

Cluster controller seed variance. The cluster two-stage controller exhibits high seed-to-seed variance ($\sigma \approx 0.4$ on key retention) because the oracle sits on a sharp 4-D feasibility boundary and the 250-cluster dataset is small. Larger datasets and feasibility-aware training losses (e.g., margin-penalized regression) are the natural next step.

Synthetic vs real TLE. The Walker shell reproduces the coverage statistics of the operational Starlink constellation but is not a snapshot of a specific real epoch. Loading a fresh Celestrak dump (`-tle PATH`) anchors the experiment to a specific UTC instant and is the figure that referees will check; the code path is identical.

Finite-key analysis. The reported rates are asymptotic; a finite-key analysis with composable security bounds, for example the tight bounds of Tomamichel *et al.* [34], is left to future work. We expect finite-key penalties to favor the adaptive controller further, since the penalty is heaviest where the sift count is smallest, exactly the regime in which the static baseline collapses.

Single-station ground segment. The Walker headline is reported for a single ground station (PTIT/Hanoi). Multi-station handover (the obvious next-decade deployment scenario for satellite QKD, projected into trusted-node-free metropolitan QKD networks such as [35] and the long-term quantum-internet vision [36]) would soften the static-baseline collapse but is beyond the scope of this paper.

C. Future Work

Three directions are immediate. (i) Generalization sweep across visibility, turbulence outer-scale, Eve distance, and ground-station latitude, since the existing scripts are parametric in all four. (ii) Finite-key correction integrated into the

per-step objective R_f ; this would tighten the secure-window estimate and likely favor the adaptive controller further, since finite-key penalties are heaviest at low sift. (iii) Multi-station fusion at the cluster head, treating the served-ground-segment trace as a partially-observed Markov decision process and re-deriving the controller as a constrained reinforcement learning agent. The current supervised pipeline already produces the oracle labels needed for an offline-RL warm start.

D. Conclusion

We have formulated, decomposed, solved, and learned a controller for the full multi-user joint optimization problem of GEO/multi-LEO/ N -user satellite QKD with SIM-BPSK/DT-DD. Validated against an SGP4-propagated Walker constellation matching Starlink density over a Hanoi ground station, the adaptive controller delivers a $2.10\times$ secret-key gain and a $3.6\times$ secure pair-time improvement, with a closed-form rule for the dual-threshold coefficient extracted from a Kolmogorov–Arnold Network. The static-parameter operating point assumed in the preceding QKD literature is shown to be a special case that fails entirely under realistic orbit geometry, repositioning parameter adaptation from an optional optimization to a system-level requirement.

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